



Prospects for Detecting Prompt X-ray Emission from Compact Object Mergers Revealed by Einstein Probe

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Collaborators:

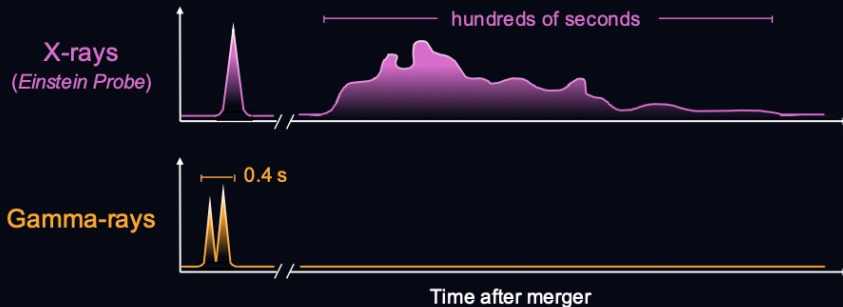
Run-Duo Liang (NAOC),

Weimin Yuan (NAOC),

and Bing Zhang (HKIAA, HKU)

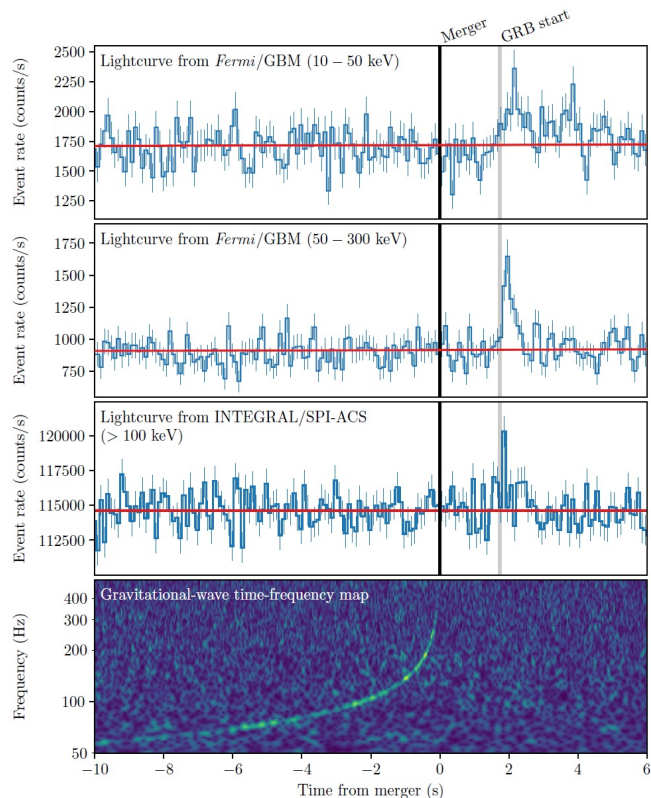


EP250704a

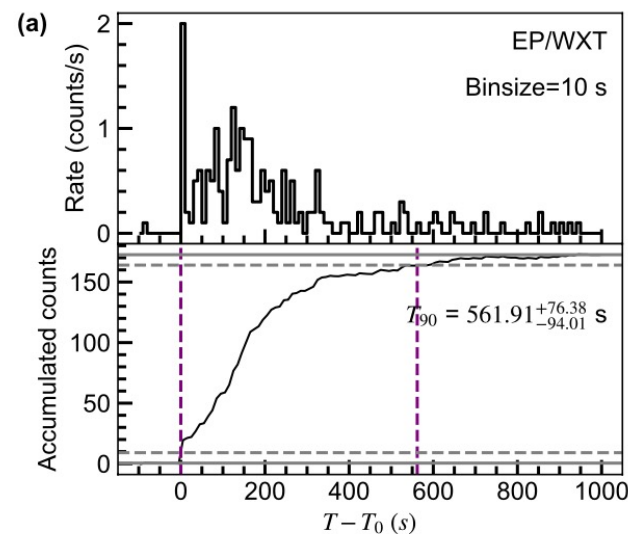
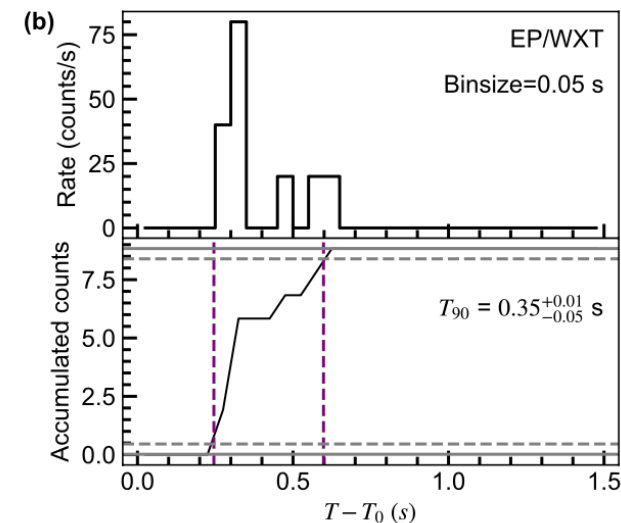
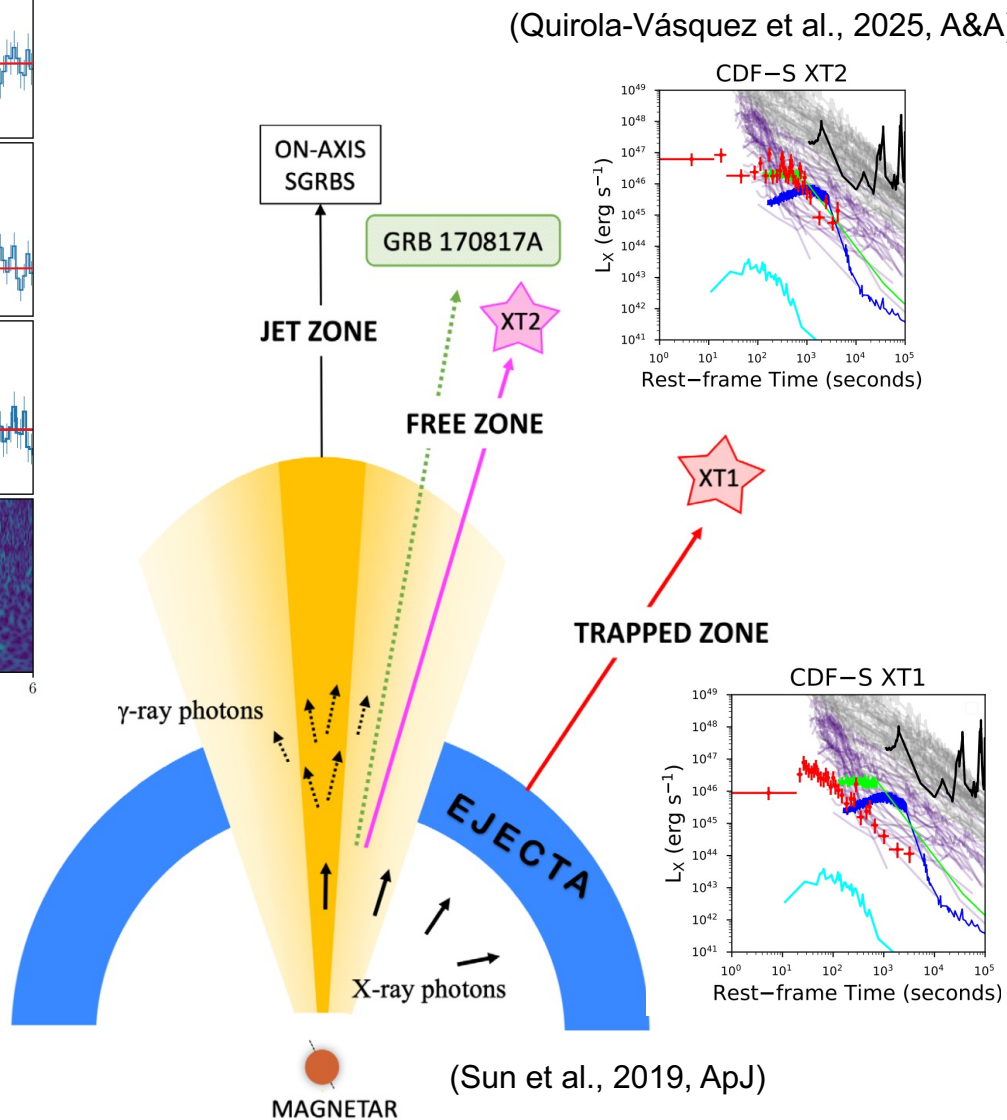




Compact object merger counterpart

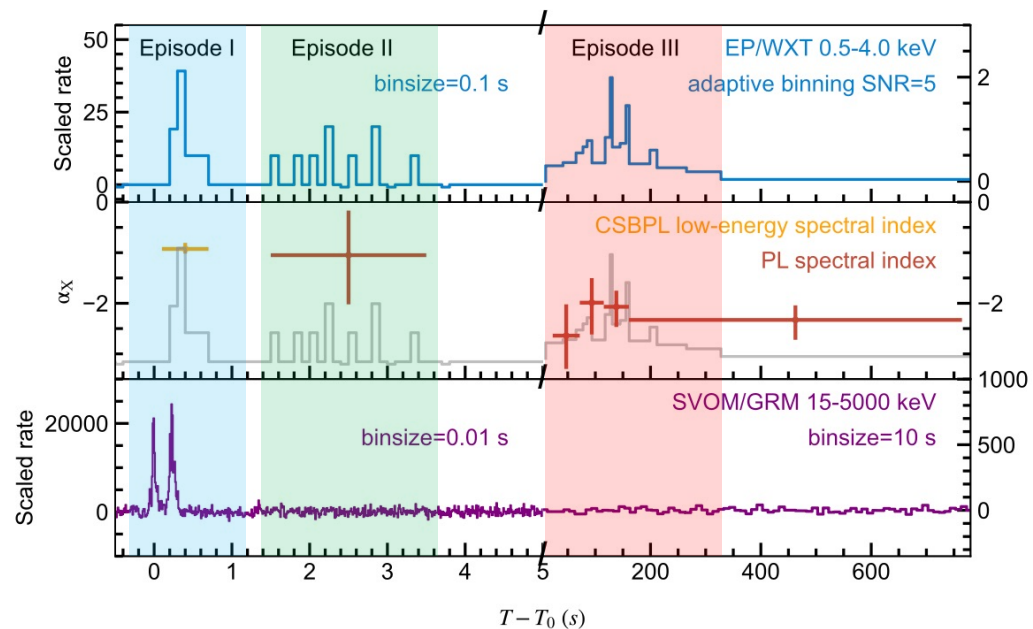


GRB 170817A and GW170817
(Abbott et al., 2017, ApJL)



EP250704a/GRB 250704B
(Li et al., 2026, SciBull in press)

First compact object merger event detected by EP

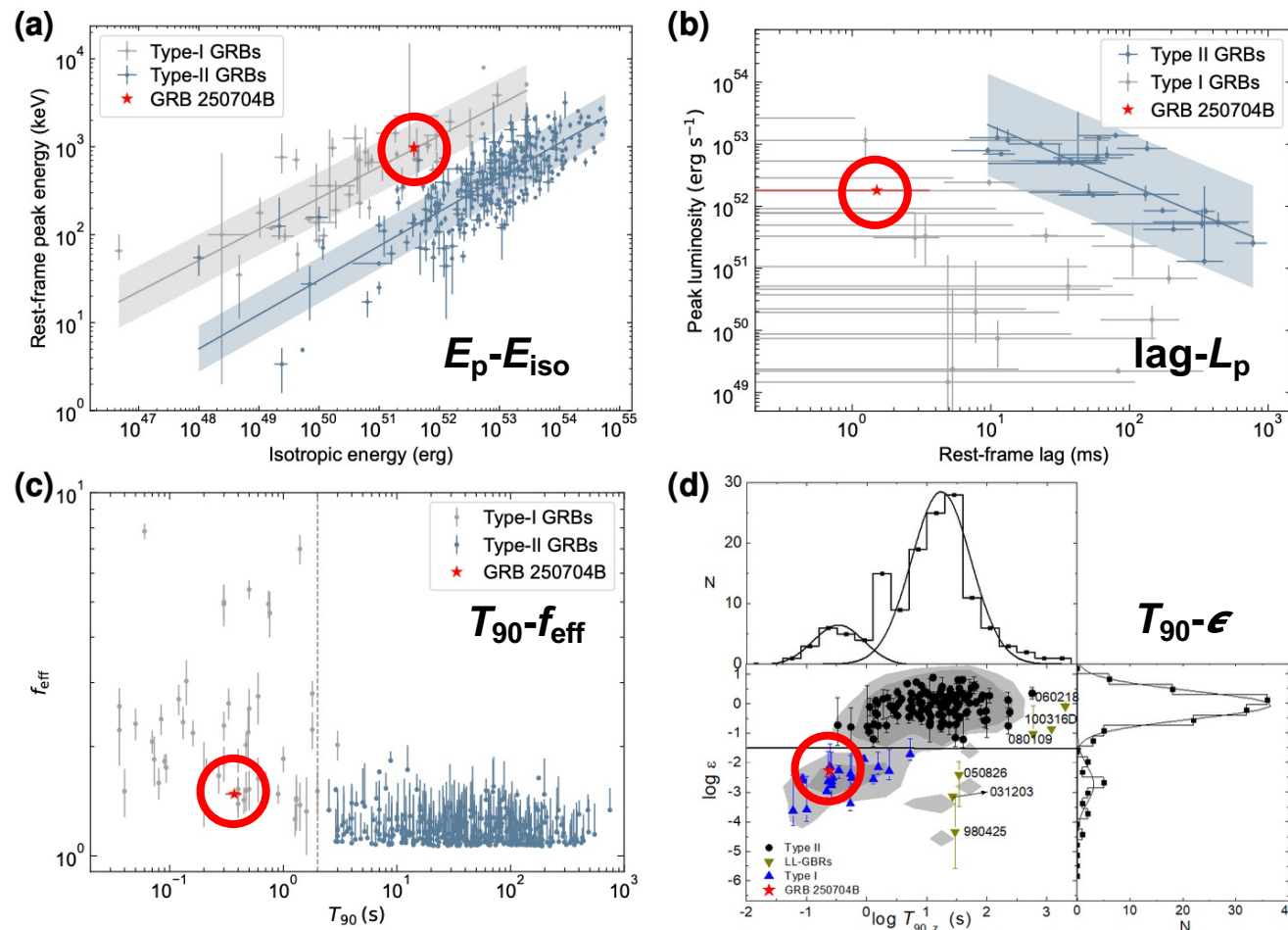


Episode I
(~ 0.5 s)
initial X-ray hard spike
gamma-ray double spikes

Episode II
(~ 3 s)
extended X-ray hard bump
no gamma-ray

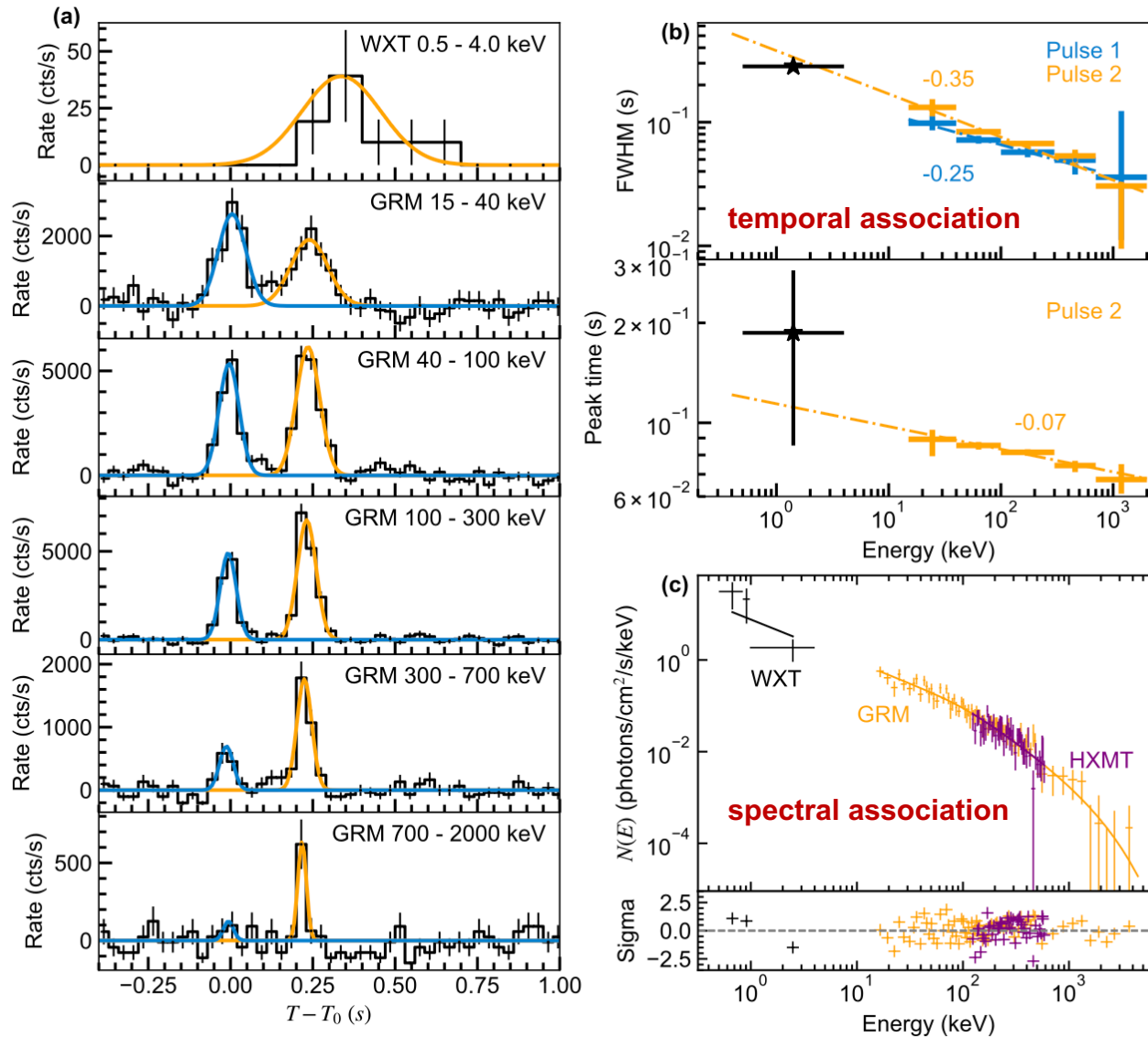
Episode III
(~ 540 s)
X-ray soft bump
no gamma-ray

GRB 250704B: a typical short GRB from merger*



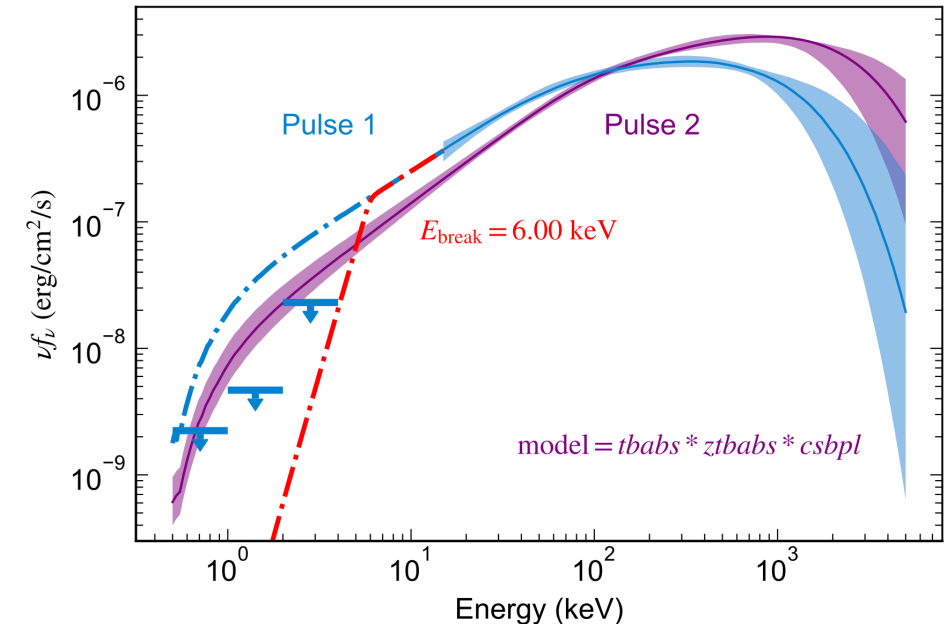
*redshift of $z=0.661$ and deep optical limits rule out a supernova

Features of the initial spike

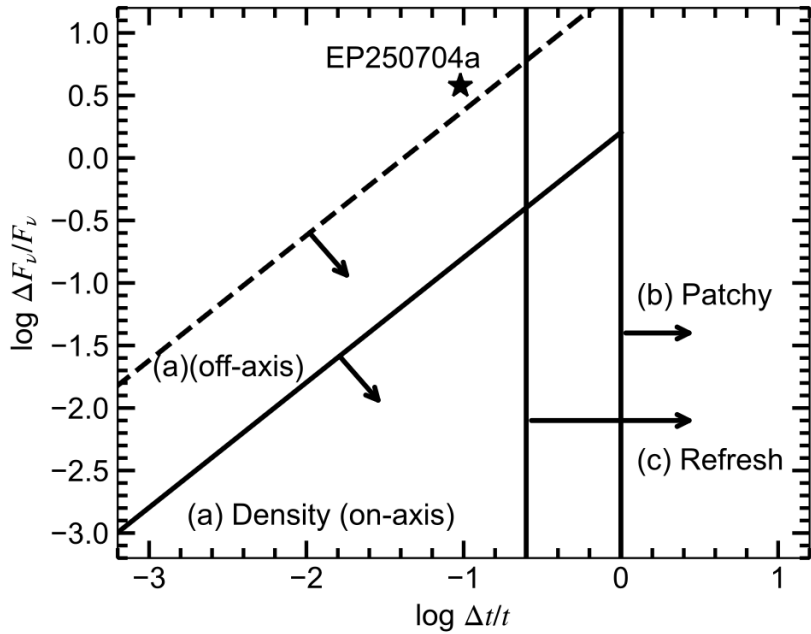


The missing X-ray initial pulse:

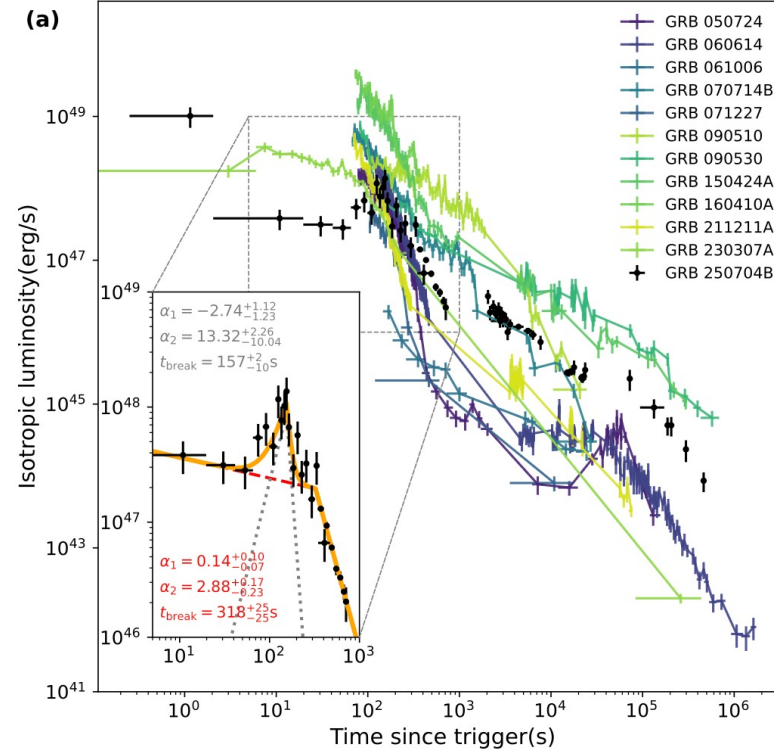
- predicted WXT counts are above detection threshold (~ 13 cts)
- a spectral break is invoked to explain the lack of Pulse 1 in X-ray band



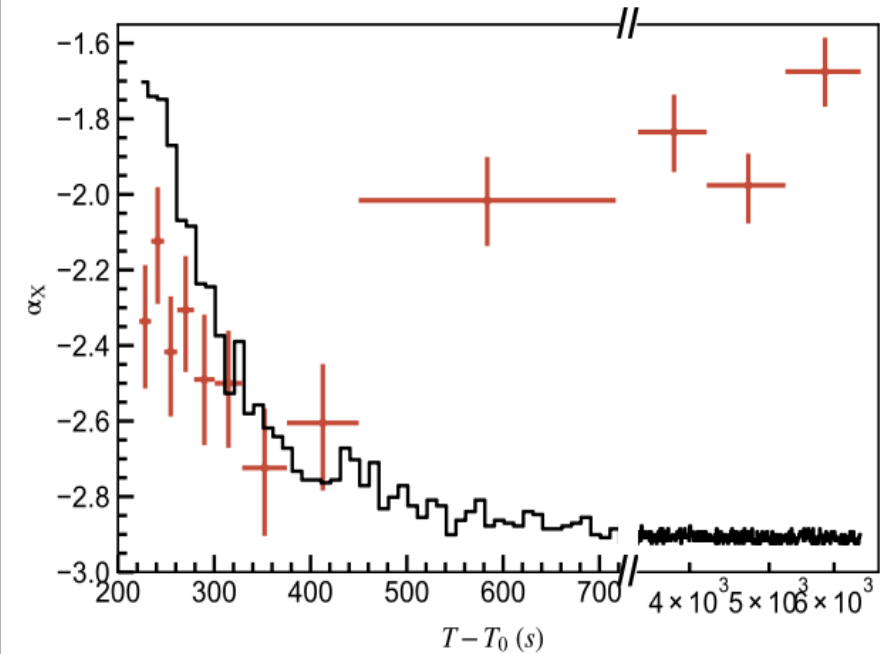
Features of the extended emission (EE)



1. rapid variability ($\Delta t/t \lesssim 0.1$, $\Delta F_v/F_v \gtrsim 3$)



2. internal plateau + steep decay

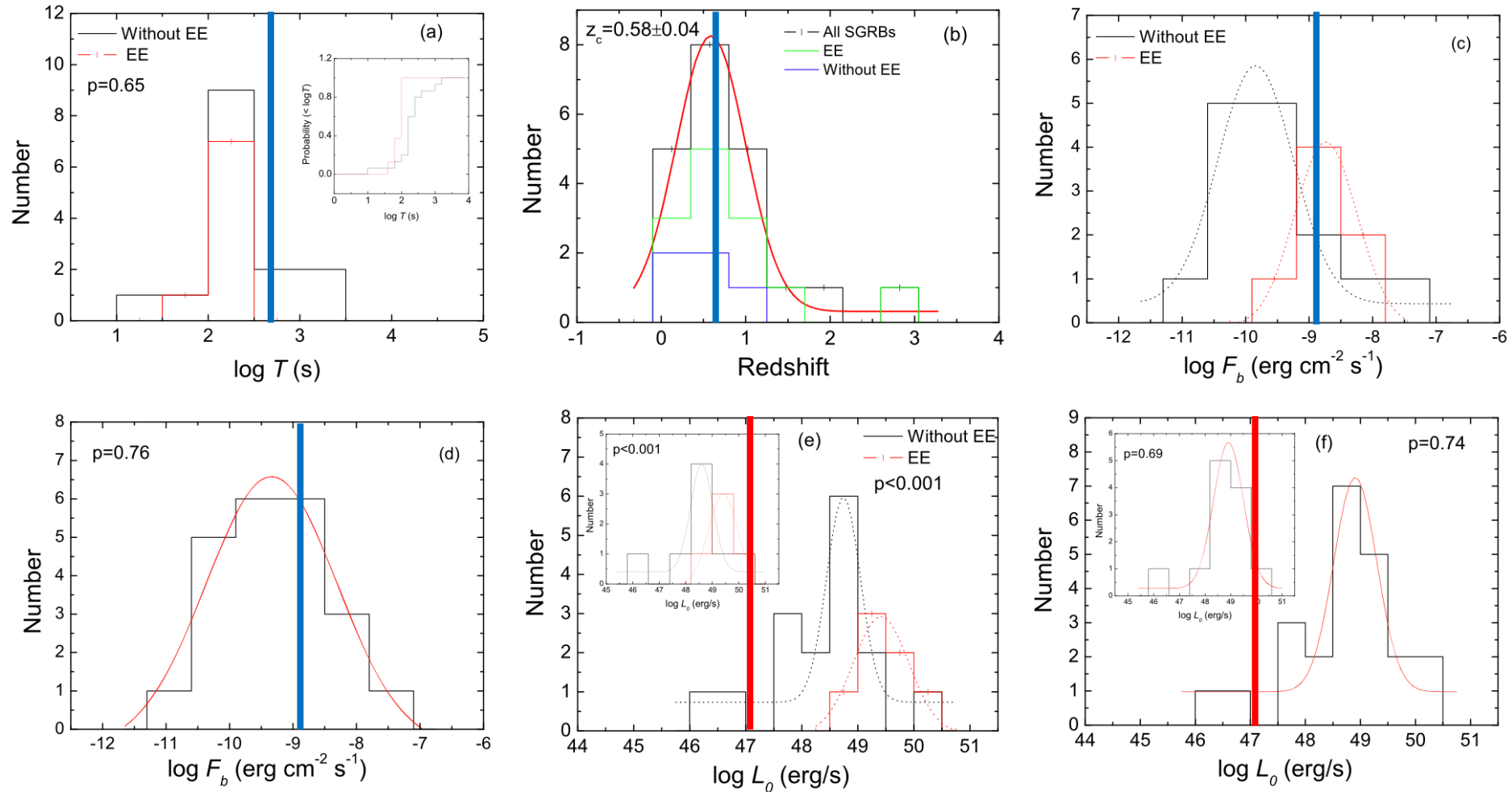


3. FXT spectra displaying evolving pattern

Temporal and spectral features support an internal origin.

EP250704a in Swift internal plateau sample

Swift-BAT detected short GRB EEs show features that are fundamentally the same as the internal X-ray plateau.
EP250704a has the average T , z , flux, but low L . $\longrightarrow$ For same L , BAT and WXT flux would be different.

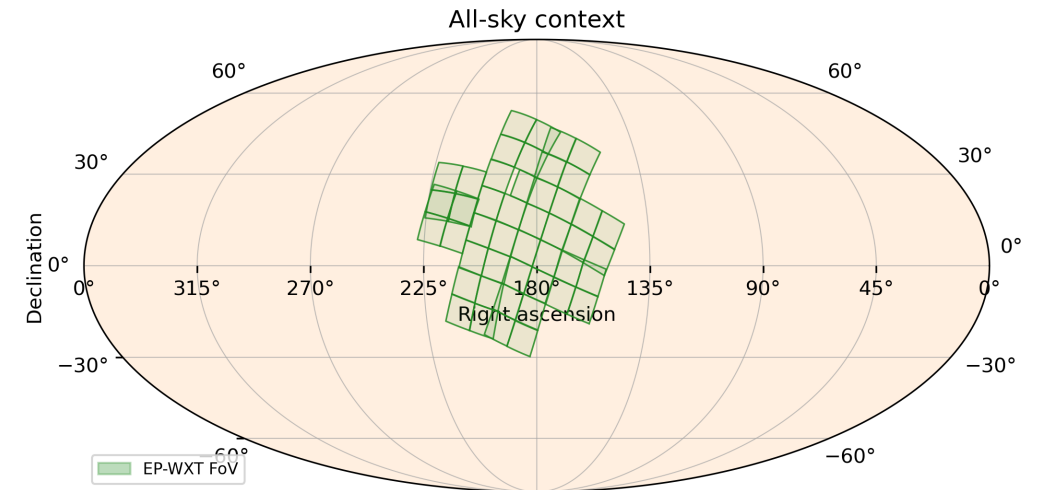




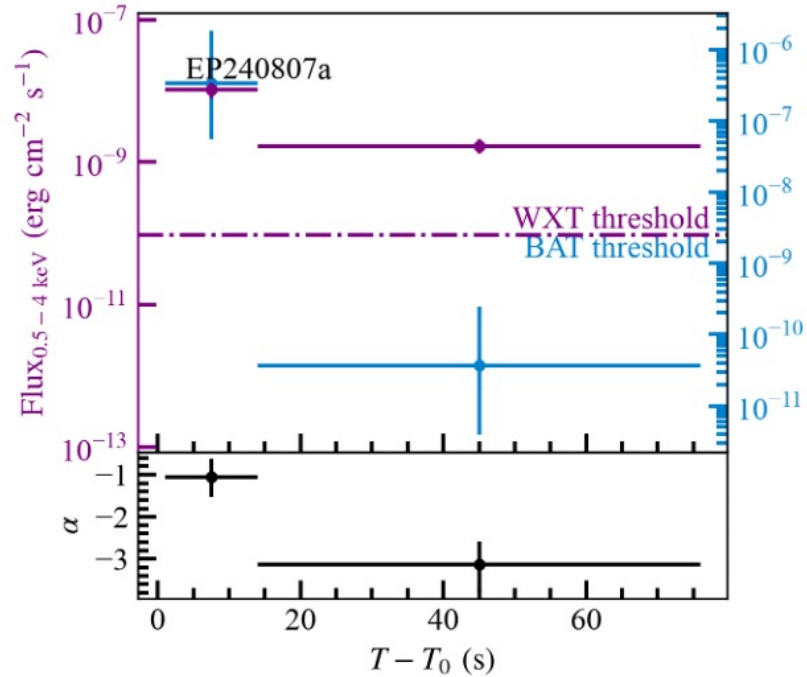
A systematic search of short GRB counterparts with EP-WXT

- **166** short GRBs from 2024-01-09 to 2026-06-01 reported in GCN
- **33** well-localized short GRBs, **3** are covered by EP within $T_0 -300/+600$ s
GRB 240807A / EP240807a, GRB 250704B / EP250704a, GRB 260527A / EP260527a
- **133** not well-localized short GRBs, **14** have partial coverage by EP

GRB 240404A (0.8793), GRB 240916B (0.8415),
GRB 241016B (0.8614), GRB 241117A (0.3812),
GRB 250119C (0.0177), GRB 250217A (0.0491),
GRB 250308A (0.3924), GRB 250728A (0.0446),
GRB 250917A (0.9649), GRB 251009A (0.0143),
GRB 251231A (0.0003), GRB 260221B (0.0978),
GRB 260411A (0.2694), GRB 260424C (0.9193)

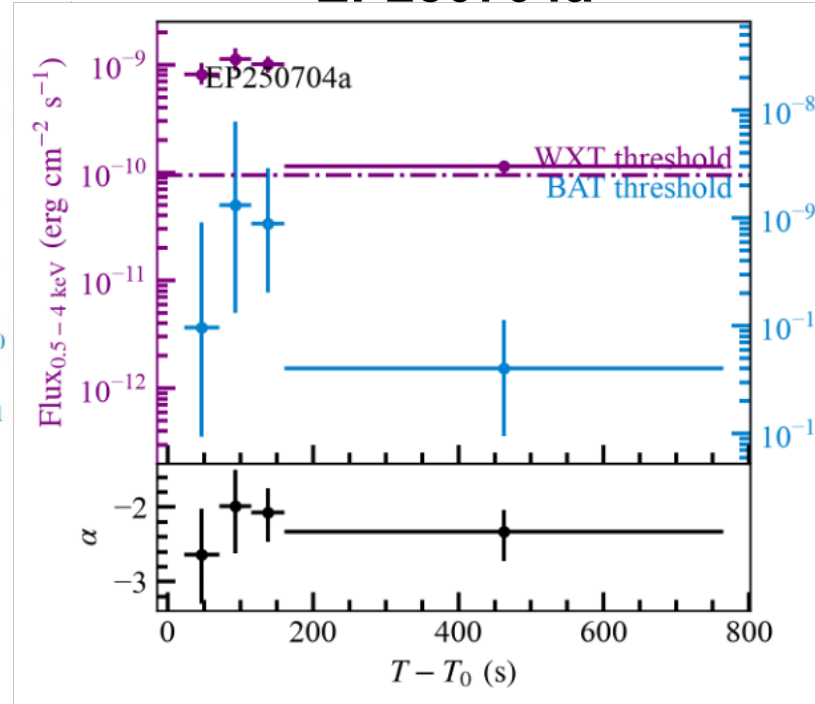


EP240807a



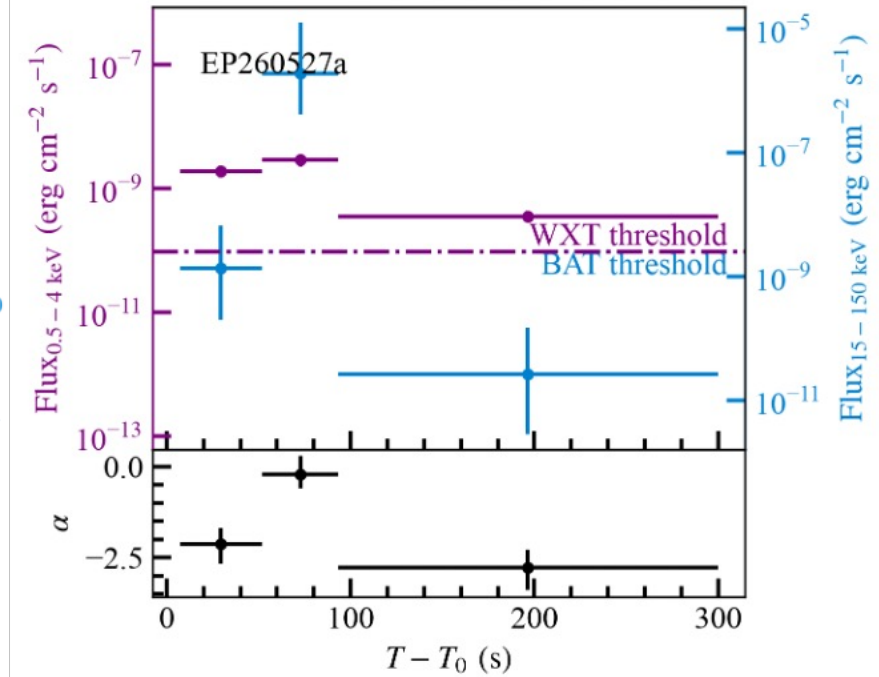
GRB duration ~ 5 s
WXT duration ~ 80 s
WXT initial spike ✗
BAT detectable? ✓

EP250704a



GRB duration ~ 0.4 s
WXT duration ~ 560 s
WXT initial spike ✓
BAT detectable? ✗

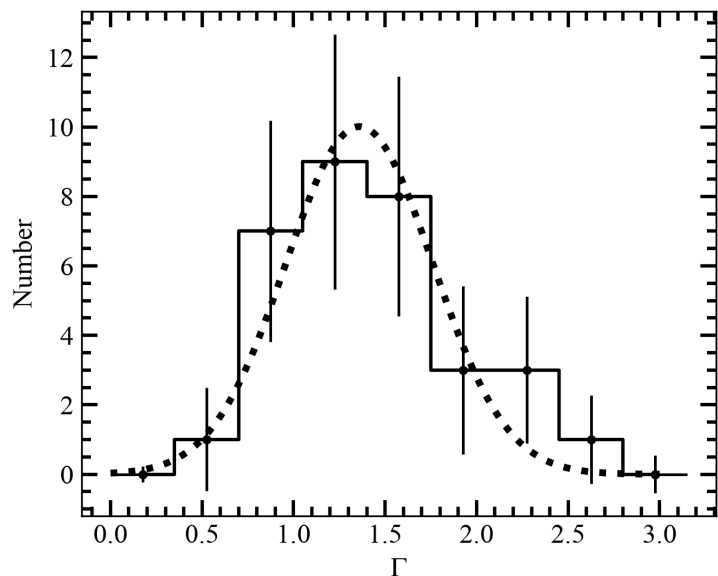
EP260527a



GRB duration ~ 0.3 s
WXT duration ~ 300 s
WXT initial spike ✗
BAT detectable? ✓

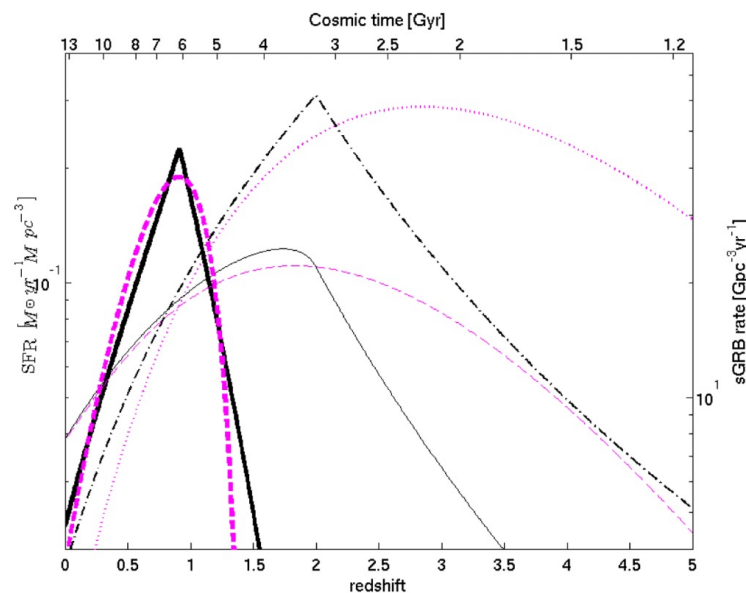
BAT detectability of WXT EE is largely dependent on spectral index.

Extended emission/internal plateau distributions



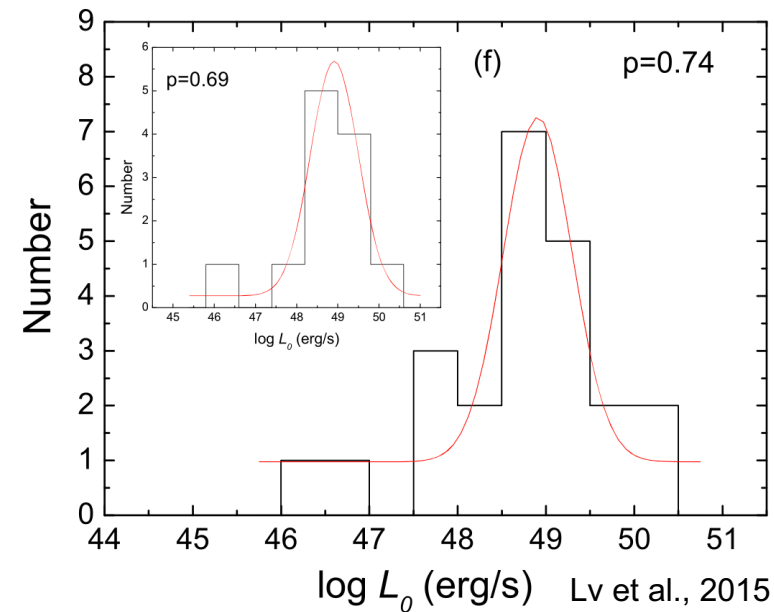
internal plateau samples (EE + AG)
from Swift and EP

photon index distribution



Wanderman and Piran, 2015

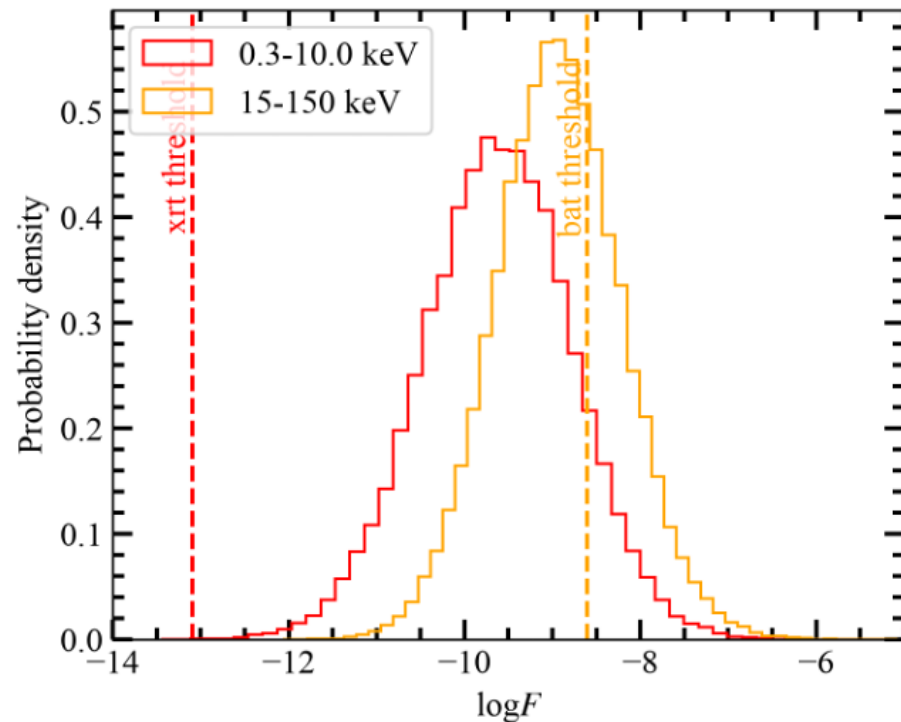
redshift distribution



Lv et al., 2015

luminosity distribution

A sanity check with Swift



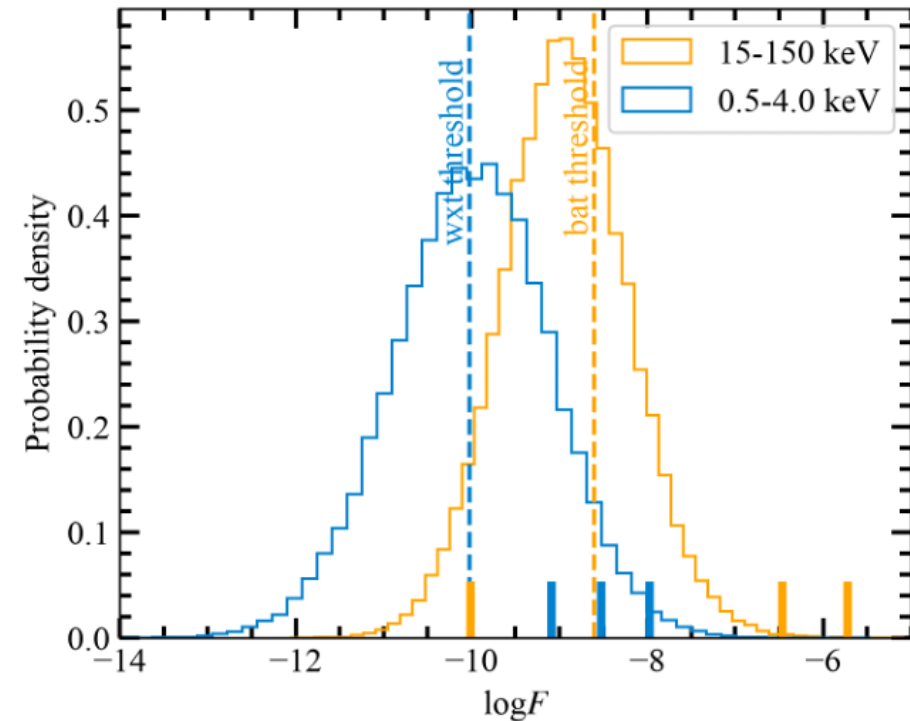
XRT IP: ~99% above threshold

BAT IP: ~31% above threshold

predicted detection rate ~ 31%

detection rate ~ 7 / 22 ~ 32%

Detection prediction on EP-WXT



BAT IP: ~31% above threshold

WXT IP: ~68% above threshold

WXT could detect relatively more extended emissions of short GRBs

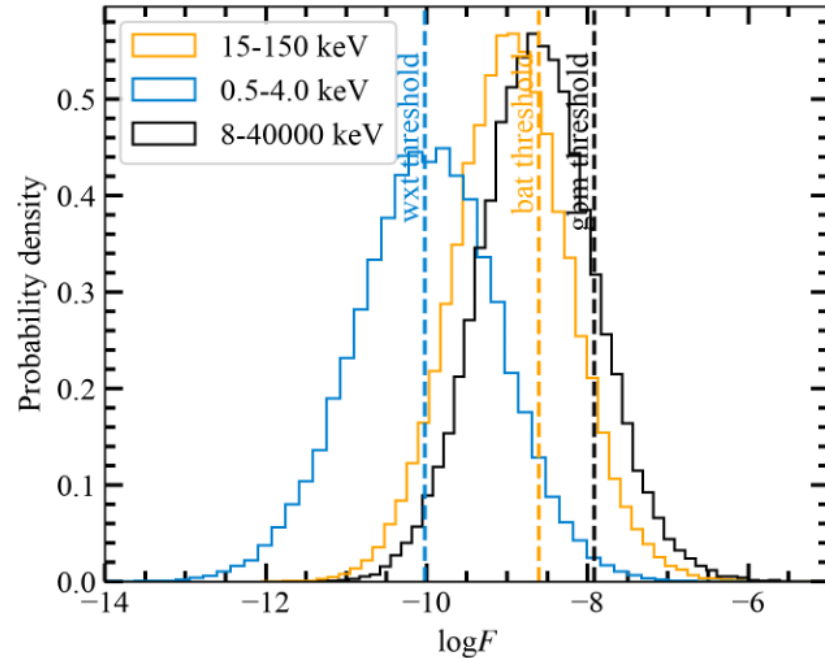
Detection prospects for EE/IP and short GRBs

EE/IP distribution

GBM: ~ 16% above threshold

BAT: ~ 31% above threshold

WXT: ~ 68% above threshold

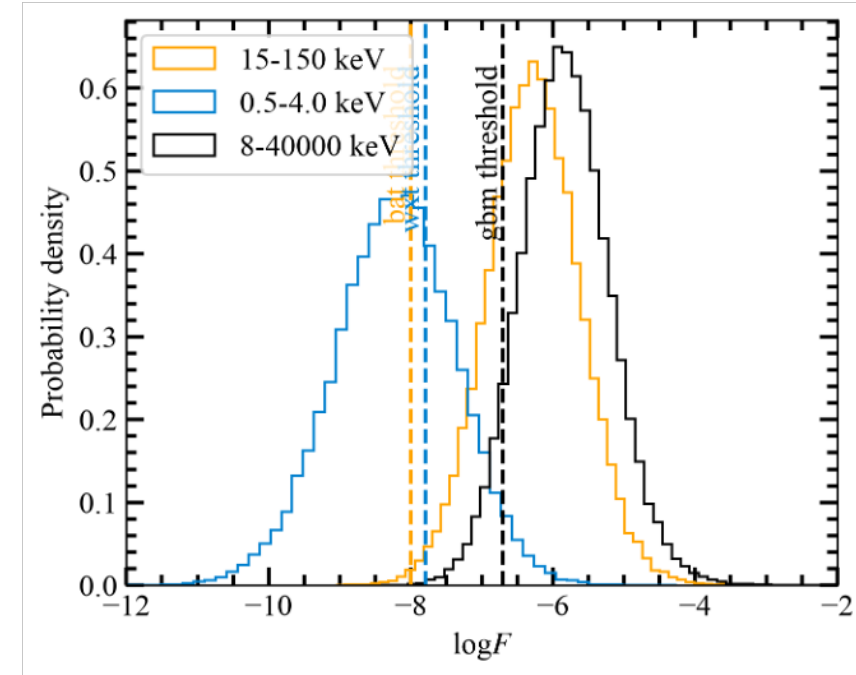


SGRB distribution

GBM: ~ 92% above threshold

BAT: ~ 100% above threshold

WXT: ~ 31% above threshold



Assuming an intrinsic fraction of N short GRBs with EE/IP,

EE/SGRB detection rate:

GBM: (16% N / 92%) = 17% N ~ 6.4% (26 / 408, Lan et al., 2020, MNRAS)

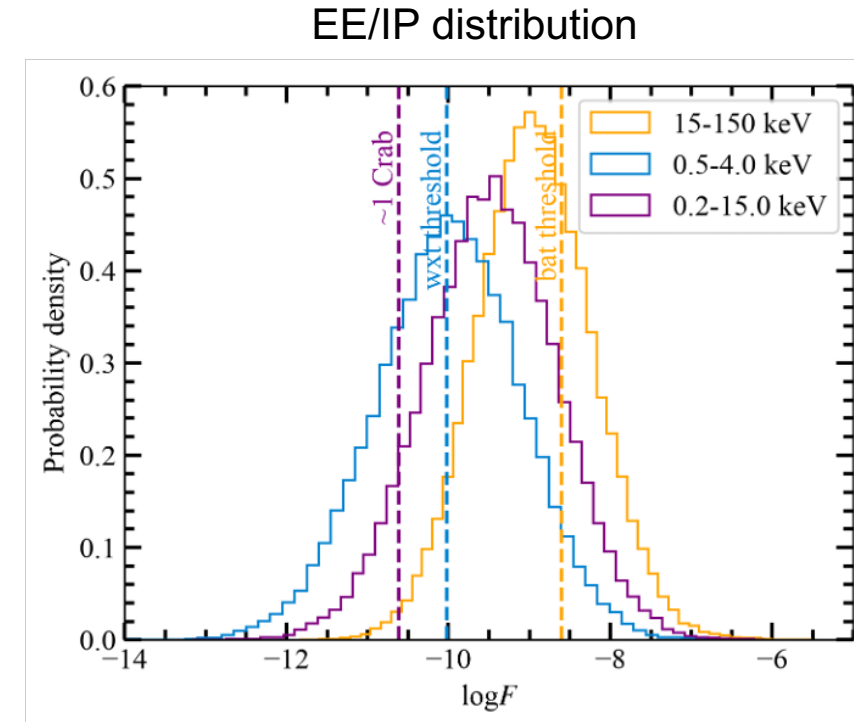
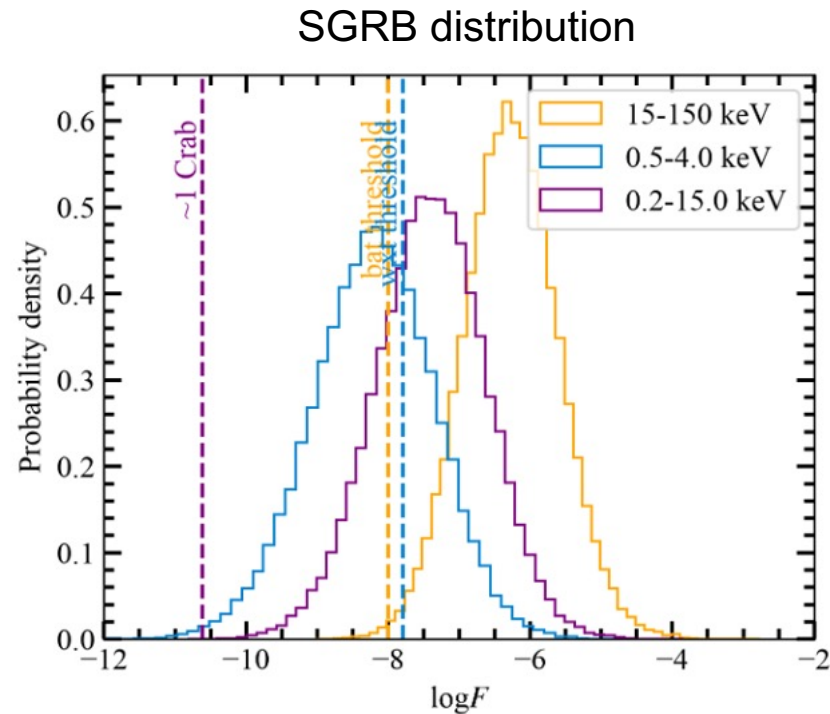
BAT: (31% N / 100%) = 31% N ~ 13.3% (12 / 90, Lien et al., 2016, ApJ)

WXT: (68% N / 31%) = 219% N ~ **83-94%**

→ **$N \sim 38-43\%$**
(~ 26-79% in Lv et al., 2015, ApJ)

Detection prospects for EE/IP and short GRBs

With New Athena Wide Field Imager:



With ~ 1 mCrab sensitivity, NewAthena could cover almost all the current population of short GRBs and $\sim 90\%$ of the extended emission/internal plateaus.

EP250704a

- **an initial spike** temporally associated with short GRB 250704B
- **a long-lasting soft X-ray internal emission** below the gamma-ray detector sensitivities
- the long-lasting X-ray emission is likely **a common feature (~40%) of merger-driven bursts**
- a promising electromagnetic counterpart to **gravitational wave sources**.

EP short GRB counterpart search

- **3 candidates** exhibiting long-lasting X-ray emission with tens to hundreds seconds duration
- the **5~14 covered non-detection** events might be intrinsic short GRBs

With New Athena

- With ~ 1 mCrab sensitivity, NewAthena could cover almost all the current population of short GRBs and around 90% of the EE/internal plateaus.

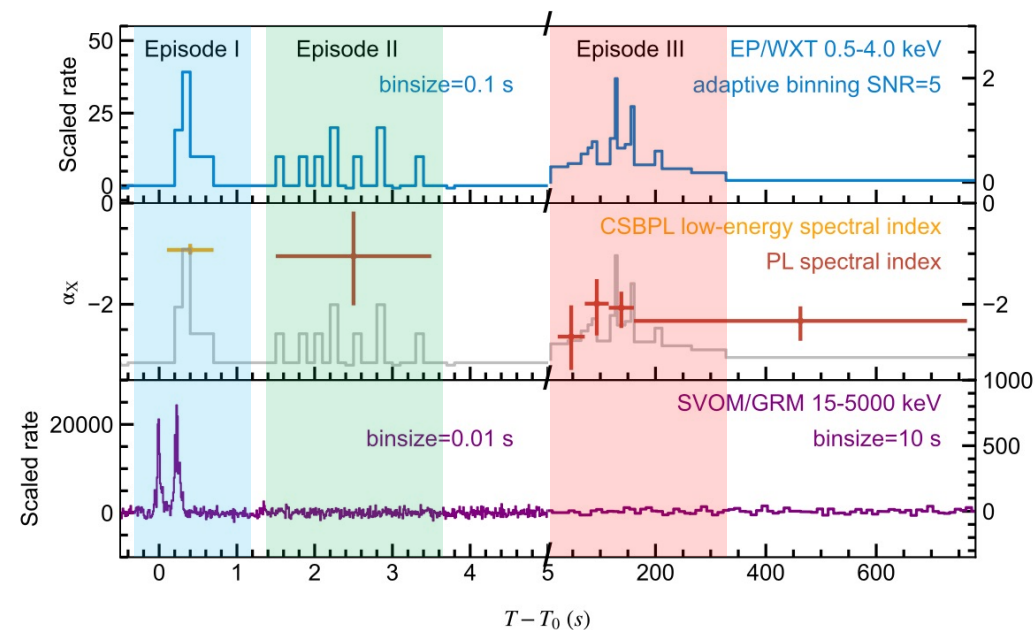


Committee on Space Research (COSPAR)



Thank you.

First compact object merger event detected by EP



Episode I

(~ 0.5 s)

initial X-ray hard spike
gamma-ray double spikes

Episode II

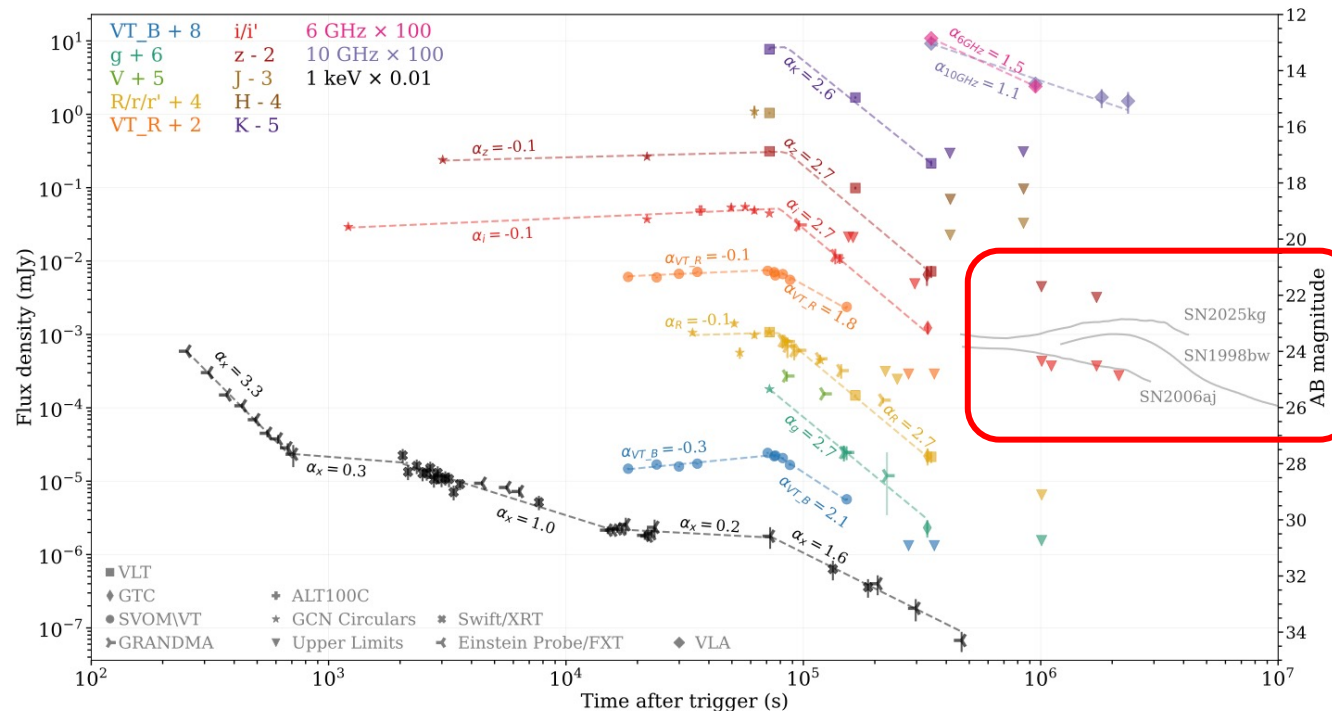
(~ 3 s)

extended X-ray hard bump
no gamma-ray

Episode III

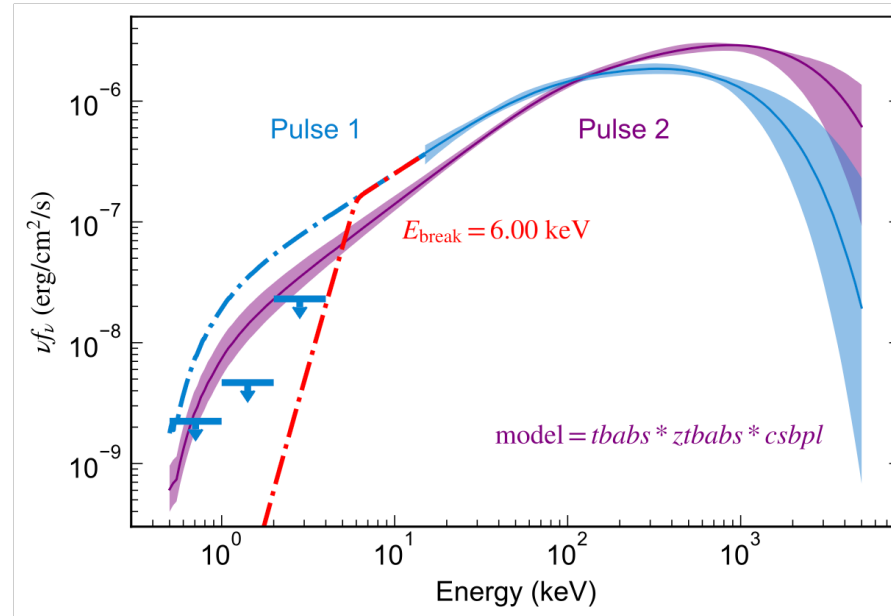
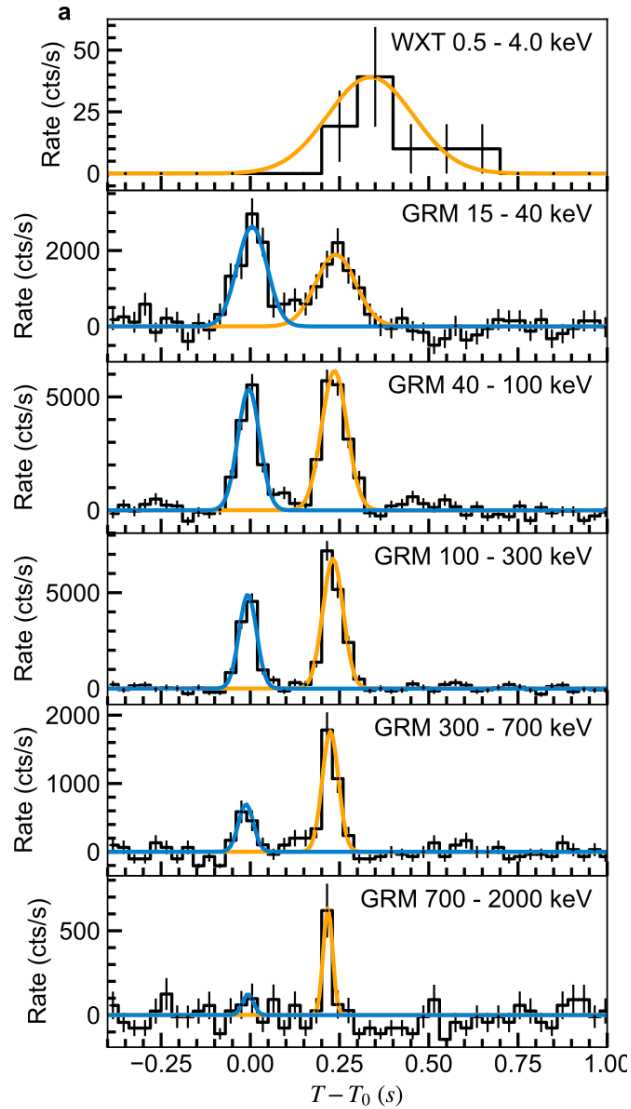
(~ 540 s)

X-ray soft bump
no gamma-ray

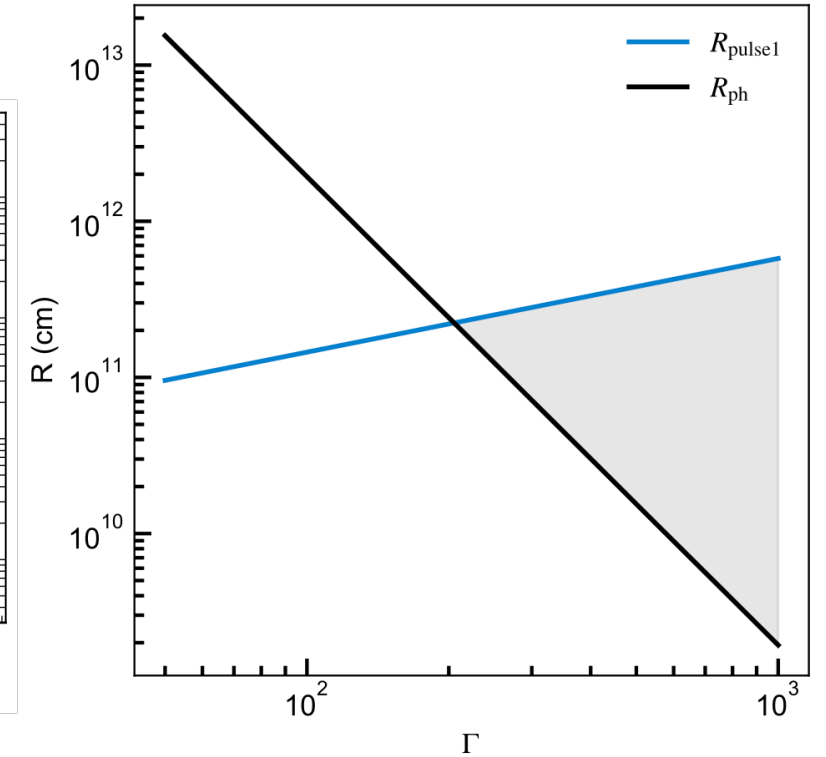


redshift of $z=0.661$ and deep optical limits rule out
a supernova

Initial spike of EP250704a / GRB 250704B



a spectral break is invoked to explain the lack of Pulse 1 in X-ray band

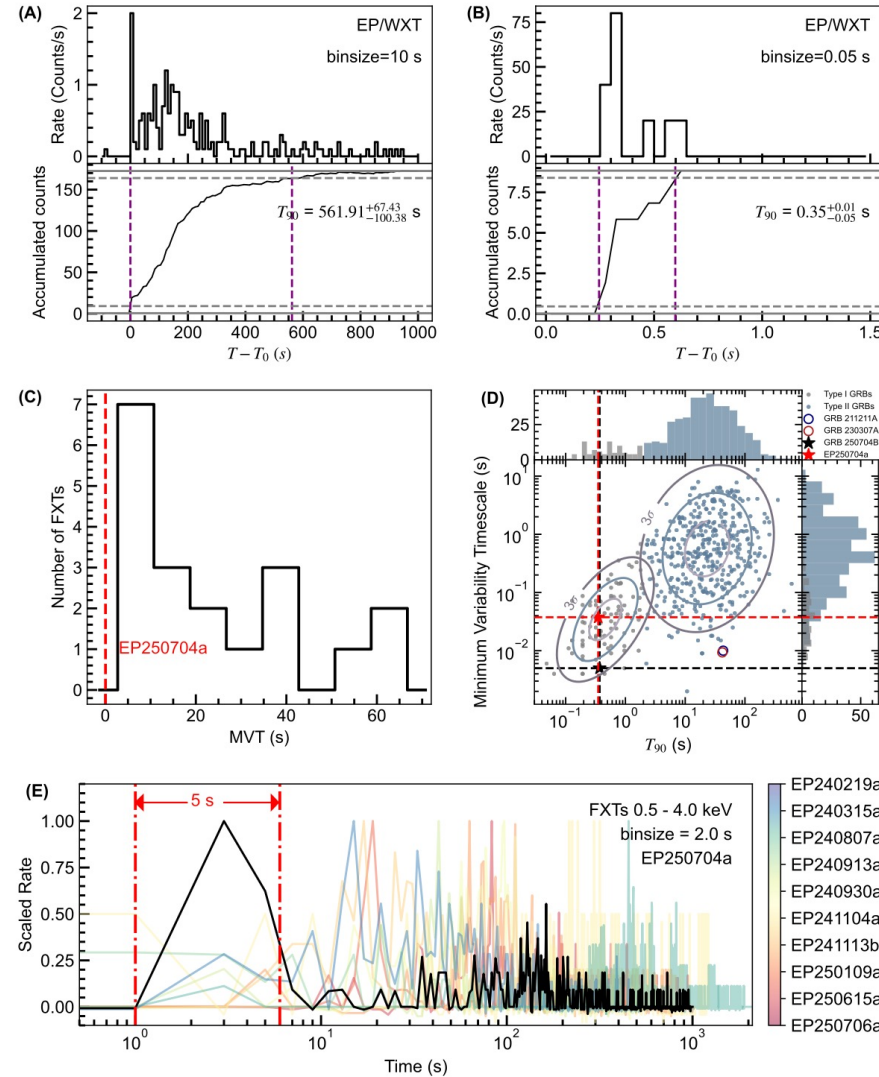




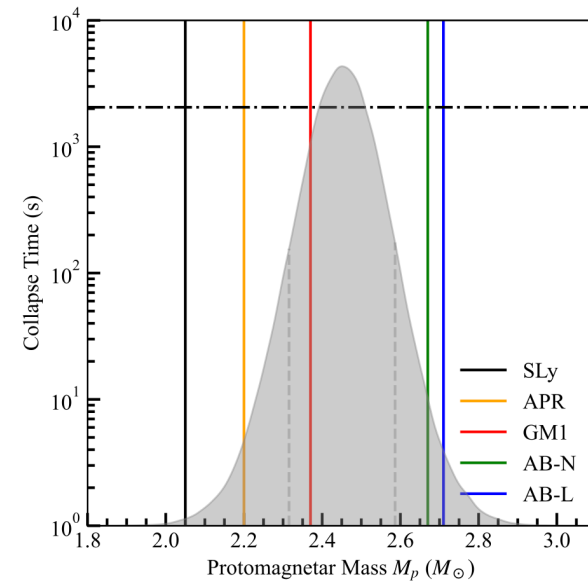
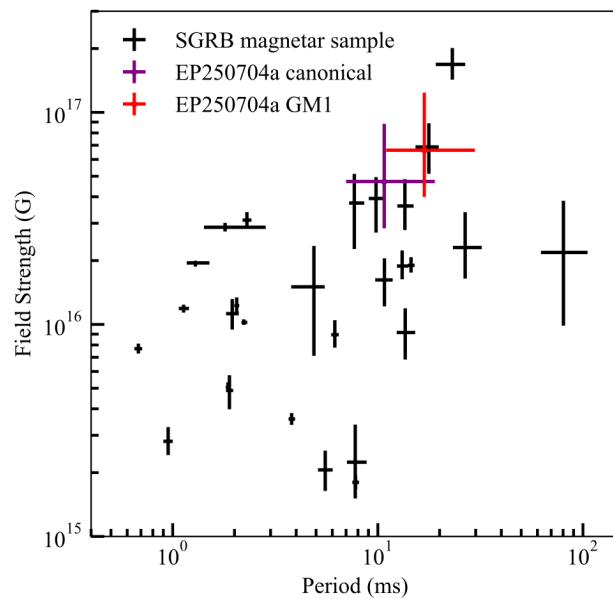
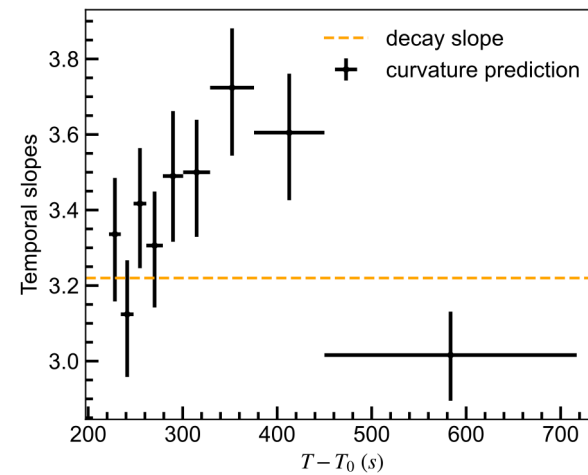
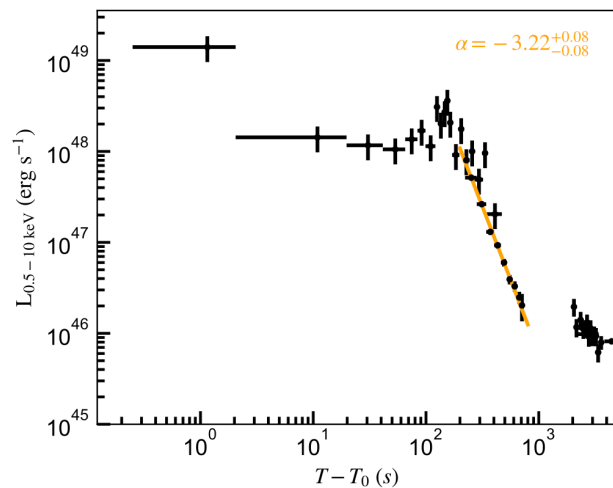
Minutes-long Soft X-ray Prompt Emission Counterpart



Among fast X-ray transient samples:



The Magnetar Central Engine

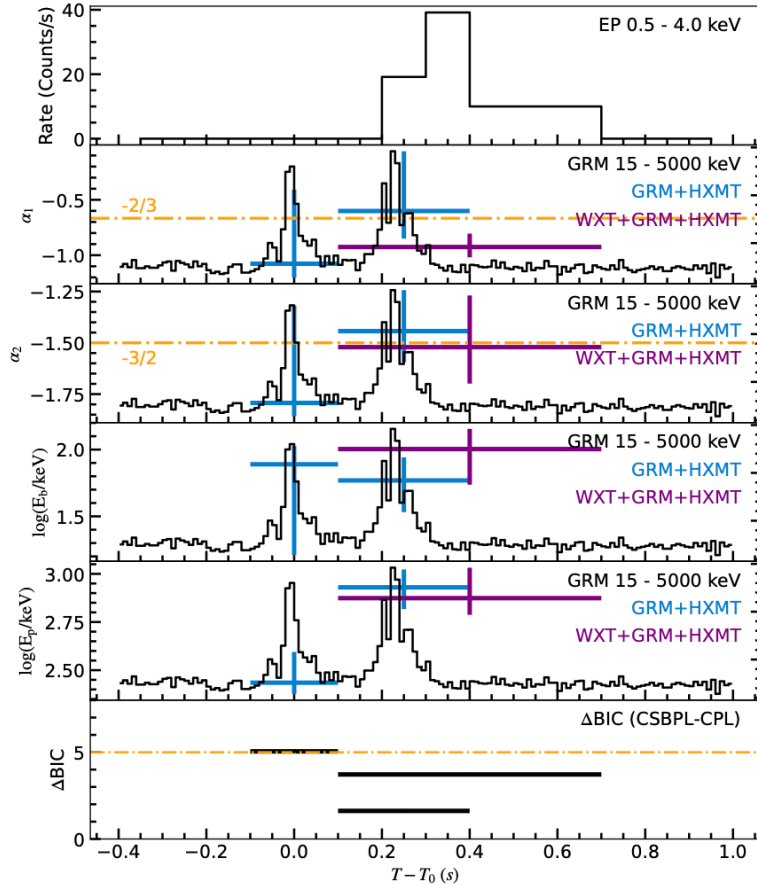




Spectral Fittings of EP250704a / GRB 250704B



t1 (s)	t2 (s)	CPL model					CSBPL model						
		α	E_p	$\log A$	stat/dof	BIC	E_b	E_p	α_1	α_2	$\log A$	stat/dof	BIC
			(photons cm ⁻² s ⁻¹ keV ⁻¹)				(keV)	(keV)			(photons cm ⁻² s ⁻¹ keV ⁻¹)		
-0.10	0.10	-1.27 ^{+0.10} _{-0.09}	334.97 ^{+43.48} _{-35.05}	-0.96 ^{+0.04} _{-0.04}	160.08/87	173.58	77.62 ^{+28.54} _{-61.41}	272.27 ^{+121.28} _{-34.04}	-1.08 ^{+0.67} _{-0.12}	-1.79 ^{+0.47} _{-0.07}	1.28 ^{+0.21} _{-0.77}	156.13/85	178.63
0.10	0.40	-1.08 ^{+0.07} _{-0.08}	626.61 ^{+87.88} _{-59.07}	-1.04 ^{+0.03} _{-0.03}	190.26/120	204.69	58.75 ^{+28.75} _{-24.79}	851.14 ^{+203.25} _{-194.99}	-0.60 ^{+0.54} _{-0.25}	-1.44 ^{+0.20} _{-0.15}	0.38 ^{+0.39} _{-0.73}	182.25/118	206.31
-0.10	0.40	-1.13 ^{+0.06} _{-0.06}	489.78 ^{+57.24} _{-35.84}	-0.99 ^{+0.02} _{-0.03}	163.85/128	178.48	55.85 ^{+57.13} _{-26.47}	588.84 ^{+91.93} _{-122.18}	-0.78 ^{+0.31} _{-0.27}	-1.46 ^{+0.22} _{-0.21}	0.75 ^{+0.40} _{-0.41}	156.73/126	181.11
0.10	0.70	-1.05 ^{+0.06} _{-0.05}	613.76 ^{+59.21} _{-59.14}	-1.04 ^{+0.03} _{-0.03}	194.83/123	209.34	100.93 ^{+41.96} _{-46.35}	748.17 ^{+328.30} _{-135.82}	-0.93 ^{+0.12} _{-0.09}	-1.52 ^{+0.25} _{-0.18}	0.88 ^{+0.15} _{-0.17}	188.88/121	213.06



t1 (s)	t2 (s)	PL Model			
		α_X	log A (photons cm ⁻² s ⁻¹ keV ⁻¹)	pgstat/dof	BIC
1.50	3.50	$-1.05^{+0.88}_{-0.97}$	$-1.80^{+1.61}_{-1.85}$	0.19/1	2.39
22.76	71.77	$-2.64^{+0.62}_{-0.66}$	$-5.83^{+1.22}_{-1.36}$	4.28/4	7.87
71.77	115.29	$-1.99^{+0.49}_{-0.63}$	$-4.45^{+0.92}_{-1.26}$	5.87/3	9.08
115.29	160.80	$-2.07^{+0.32}_{-0.40}$	$-4.64^{+0.62}_{-0.78}$	12.28/10	17.25
160.80	764.50	$-2.33^{+0.29}_{-0.39}$	$-6.08^{+0.56}_{-0.78}$	11.11/16	16.89
22.76	764.50	$-2.30^{+0.23}_{-0.23}$	$-5.75^{+0.46}_{-0.46}$	174.26/39	181.68
222.00	234.50	$-2.34^{+0.15}_{-0.18}$	$-5.65^{+0.28}_{-0.34}$	58.83/57	66.98
234.50	248.00	$-2.12^{+0.14}_{-0.17}$	$-5.29^{+0.27}_{-0.31}$	49.25/58	57.43
248.00	261.50	$-2.42^{+0.15}_{-0.17}$	$-5.86^{+0.28}_{-0.32}$	51.46/56	59.58
261.50	279.00	$-2.31^{+0.14}_{-0.16}$	$-5.75^{+0.26}_{-0.30}$	40.87/55	48.96
279.00	300.50	$-2.49^{+0.17}_{-0.17}$	$-6.21^{+0.33}_{-0.34}$	45.98/56	54.1
300.50	329.00	$-2.50^{+0.14}_{-0.17}$	$-6.34^{+0.25}_{-0.33}$	62.13/56	70.25
329.00	375.50	$-2.72^{+0.16}_{-0.18}$	$-6.97^{+0.30}_{-0.35}$	49.12/56	57.25
375.50	450.00	$-2.60^{+0.16}_{-0.18}$	$-7.12^{+0.30}_{-0.36}$	68.94/55	77.03
450.00	717.00	$-2.02^{+0.11}_{-0.12}$	$-6.36^{+0.21}_{-0.24}$	65.42/78	74.18
3460.00	4196.50	$-1.83^{+0.10}_{-0.11}$	$-6.67^{+0.18}_{-0.20}$	122.06/107	131.44
4196.50	5234.50	$-1.98^{+0.08}_{-0.10}$	$-7.00^{+0.16}_{-0.20}$	129.85/126	139.55
5234.50	6464.00	$-1.68^{+0.09}_{-0.09}$	$-6.45^{+0.17}_{-0.18}$	151.30/145	161.28